Store Management System — Class Diagram

List-only · .txt persistence · console · no Exception (Result<T> / boolean)

Legend

- Application (Console UI)
- Services
- Result<T>
- Persistence (.txt)
- Domain: Parties, Menu & Inventory, Transactions

solid arrow = association · dashed arrow = dependency
hollow triangle = inheritance (dashed = implements)
filled diamond = composition

Main
+main(args: String[]): void

ConsoleApp
+run(): void
-showMainMenu(): void

CrudService<T>
#repository: Repository<T>
+add(entity: T): Result<T>
+update(entity: T): Result<T>
+deactivate(id: String): Result<T>
+getById(id: String): Result<T>
+getAll(): List<T>
#{abstract} newId(): String

«utility»
ConsoleInput
+readRequiredText(msg: String): String
+readPositiveInt(msg: String): int
+readPositiveDouble(msg: String): double

«utility»
Validator
+isValidPhone(phone: String): boolean
+isValidEmail(email: String): boolean

OrderService
+createOrder(customerId: String, employee: Employee): Result<Order>
+addItem(orderId: String, itemId: String, qty: int): Result<Order>
+confirmOrder(orderId: String): Result<Order>
+cancelOrder(orderId: String): Result<Order>

ImportService
+createImport(supplierId: String, employee: Employee): Result<ImportOrder>
+addItem(importId: String, itemId: String, qty: int, cost: double): Result<ImportOrder>
+receiveImport(importId: String): Result<ImportOrder>

SupplierService

MenuService

CustomerService

EmployeeService

«utility»
IdGenerator
+next(prefix: String, existingIds: List<String>): String

AuthService
+login(username: String, password: String): Result<Employee>

InventoryService
+receive(item: MenuItem, qty: int, ref: String, by: Employee): Result<StockEntry>
+issue(item: MenuItem, qty: int, ref: String, by: Employee): Result<StockEntry>
+adjust(item: MenuItem, newQty: int, by: Employee): Result<StockEntry>
+getQuantity(itemId: String): int
+getHistory(itemId: String): List<StockLog>

ReportService
+revenueBetween(from: LocalDate, to: LocalDate): double
+lowStockItems(threshold: int): List<MenuItem>

Result<T>
-success: boolean
-message: String
-data: T
+{static} ok(data: T): Result<T>
+{static} fail(message: String): Result<T>
+isSuccess(): boolean
+getMessage(): String
+getData(): T

«interface»
Repository<T>
+add(entity: T): void
+update(entity: T): void
+findById(id: String): T
+findAll(): List<T>

TextFileRepository<T>
-filePath: String
-items: List<T>
-converter: TextConverter<T>
+load(): void
+flush(): void

«interface»
TextConverter<T>
+toLine(entity: T): String
+fromLine(line: String): T

«utility»
FileHandler
+readLines(path: String): List<String>
+writeLines(path: String, lines: List<String>): void

CustomerConverter

EmployeeConverter

SupplierConverter

MenuItemConverter

OrderConverter

ImportOrderConverter

StockEntryConverter

StockLogConverter

BaseEntity

Transaction<I>
-time: LocalDateTime
-status: TransactionStatus
-items: List<I>
+addItem(item: I): boolean
+removeItem(itemId: String): boolean
+getItems(): List<I>
+getTotal(): double
+confirm(): boolean
+cancel(): boolean
#{abstract} onConfirm(): void

Party
-name: String
-phoneNumber: String
-address: String
+rename(name: String): void
+changeContact(phone: String, address: String): void

StockEntry
-quantity: int
+increase(n: int): boolean
+decrease(n: int): boolean
+getQuantity(): int

StockLog
-logTime: LocalDateTime
-changeType: StockChangeType
-quantity: int
-quantityBefore: int
-quantityAfter: int
-refId: String

«factory»
MenuItemFactory
+create(type: String, fields: List<String>): MenuItem

«interface»
IPayable
+calculatePay(): double

«enumeration»
TransactionStatus
DRAFT
CONFIRMED
COMPLETED
CANCELLED

LineItem
-quantity: int
-unitAmount: double
+getSubTotal(): double
+changeQuantity(q: int): boolean

Person

Order
+complete(): boolean
+getFinalTotal(): double

ImportOrder

MenuItem
-name: String
-price: double
-description: String
+changePrice(newPrice: double): void

«enumeration»
StockChangeType
IMPORT
SALE
ADJUSTMENT
RETURN

Employee
-role: Role
+{abstract} calculatePay(): double

ImportItem

OrderItem

«interface»
DiscountPolicy
+apply(subtotal: double, customer: Customer): double

Customer
-loyaltyPoints: int
-tier: MemberTier
+earnPoints(amountSpent: double): void
+redeemPoints(points: int): boolean

Supplier
-companyName: String
-email: String

Drink
-size: DrinkSize

FastFood
-calories: double

«enumeration»
Role
ADMIN
MANAGER
CASHIER
WAITER
+can(p: Permission): boolean

FullTimeEmployee
-monthlySalary: double
+calculatePay(): double

PartTimeEmployee
-hourlyRate: double
-hoursWorked: double
+calculatePay(): double

Account
-username: String
-passwordHash: String
+authenticate(rawPassword: String): boolean
+changePassword(oldPassword: String, newPassword: String): boolean

MemberDiscount

NoDiscount

«enumeration»
MemberTier
NONE
SILVER
GOLD

«enumeration»
DrinkSize
S
M
L

«enumeration»
Permission
MANAGE_STAFF
MANAGE_MENU
CREATE_ORDER
IMPORT_STOCK
VIEW_REPORT